```
class Demo{
    psvm(){
    int n=4
    for(int i=1; i<=n; i++){
    for(int s=1; s<=n-i; s++){
        Sop("--");
        }
    for(int j=1; j<=i; j++){
        Sop("*");
        }
    Soln();
    }}
```

o/p →

```
   *
  * *
 * * *
* * * *
```

Time Complexity : $i \rightarrow [1-n] \rightarrow n$

$s \rightarrow [1-n-i] \rightarrow n$

$j \rightarrow [1-i] \rightarrow \frac{n+1}{2}$

$O(n^2)$.

Space Complexity → $O(1)$.

```
class Demo{
psvm(){
for(int i=1; i<=n; i++){        —— n
for(int j=1; j<=n; j++){        —— n
for(int k=1; k<=n; k++){        —— n
    Sop("*");
    }}}
```

Time Complexity : $n \times n \times n = n^3$

$O(n^3)$.

Space Complexity

$O(1)$.

Assignment No. 4

```
class Demo{
    psum(){
        int n=4
        for(int i=1; i<=n; i++){
            sop(" * ");
        }
    }
```

~~Time Compl~~

Time complexity ⟹ O(n)
Space complexity ⟹ O(1).

```
class Demo{
    psum(){
        int n=4;
    for(int i=1; i<=n; i++){
     for(int i=1; j<=n; j++){
        sop(" * ");
     }
     sopln();
    }
}
```

Time Complexity. ⇒

outer loop → n
Inner loop → n] — n×n = n^2

$O(n^2)$

Space Complexity → O(1)

```
class Demo{
    psvm(){
    int n = 48;
    for(int i = 1; i <= Math.sqrt(n); i++){
        if(n%i == 0){
            sop(i + " " + (n/i));
        }
    }
}
```

output

1	48
2	24
3	16
4	12
6	8

Time Complexity →

iteration – $\sqrt{n}$ time

$\sqrt{48} = 6.9$ → 1 2 3 4 5 6

Time Complexity → $O(\sqrt{n})$

Space Complexity ⇒ $O(1)$.

```
psvm(){
int n = 12345;
while(n>0){
    int digit = n%10;
    sop(digit);
    n = n/10;
}
```

output

5
4
3
2
1.

Time Complexity → d = no of digits ~~O(d)~~

$\boxed{O(d)}$

Space Complexity $\boxed{O(1).}$

```
class Demo{ int n=16
for(int i=n;i>=1;i=i/2){
        sop(i);
    }
```

$i=n$, $i=n/2$, $i=n/4$ $i=n/8$ ...

$n/2^k = 1$

$2^k = n$ $\Rightarrow$ $\log 2^k = \log n$

$k \log 2 = \log n$

$k = \log_2 (n)$

Time Complexity $\rightarrow$ $O(\log n)$

Space Complexity $\rightarrow$ $O(1)$.

```
class Demo{
    psvm(){
    int n=4;
    for(int i=1;i<=n;i++){          —— n
    for(int j=1;j<=n;j=j*2){        —— log n
        sop(i + " " + j);
    }
}
```

Time Complexity $\rightarrow$ $O(n \log n)$.

Space Complexity $\rightarrow$ $O(1)$.